Resonant injection along a pump-written interface: time-dependent simulations and an edge-chain analysis

Working report II, adiabatic_topology_blueshift (EnO run 2026) · 10 October 2026 · builds on Report I (stationary design, 9 October 2026)

Abstract. Report I showed that a non-resonant pump, treated as a static exciton blueshift, writes an interface into a lattice of expanded hexamers, and that the 13 edge pillars along a straight 60° pump edge carry a band of bright interface modes. The present work addresses whether this interface transports light. The static blueshift is replaced by a driven–dissipative Gross–Pitaevskii model with an incoherent exciton reservoir, gain, exciton–exciton interaction, an inhomogeneous exciton line and spontaneous-emission noise. Under a cw incoherent pump the model reproduces the interface of Report I in photoluminescence below threshold (edge-to-bulk contrast 14 at 0.81 $P_{\mathrm{th}}^{\mathrm{est}}$, Fig. 1). Resonant cw injection at the lowest edge pillar, however, does not propagate: the coherent signal drops by 1–2 orders of magnitude per hexamer and reaches the noise floor after the second one, about 9 µm along a 40 µm edge, for both interface geometries, all drive wave vectors, energies, densities, polarisations, pump powers and disorder seeds, and also for a small non-resonant spot (Figs. 2, 3). A frequency-domain transmission calculation shows the same for the stationary design itself, and in the simulated photoluminescence far field the interface emits as two flat lines (Fig. 4). The eigenmodes explain why: the interface band is an almost ideal Su–Schrieffer–Heeger chain of edge-pillar dimers, one per boundary hexamer, with intra-dimer hopping $t_1 = -59.5$ µeV and inter-dimer hopping only $t_2 = +12.0$ µeV (Fig. 5); a Bloch calculation of a supercell across the interface yields the same two isolated bands (Fig. 8). Its bands are 22–26 µeV wide, seven times narrower than the polariton linewidth, so the maximum propagation length $a_{\mathrm{latt}}|t_2|/\Gamma$ is 0.5 µm without and 4 µm with the gain of a pump at 0.9 $P_{\mathrm{th}}^{\mathrm{est}}$. Without free parameters, the chain model reproduces the first step of the simulated decay and bounds the subsequent ones from above (Fig. 6). Pushing the pump to 1.01 $P_{\mathrm{th}}^{\mathrm{est}}$, just below the lasing threshold, or raising the injected density 300-fold does not extend the reach, because the transfer between dimers becomes limited by their energy mismatch rather than by the loss (Fig. 7). Routing over the 40 µm interface requires $|t_2|/\Gamma \approx 6$, i.e. a design with a direct coupling between neighbouring edge dimers.

1 Introduction

In Report I [1] a pump spot entered the stationary polariton eigenproblem only as a local exciton shift $B_{\mathrm{eff}} = 2.61$ meV. Along a straight pump edge at 60° to the x axis, parallel to a lattice vector and placed on a boundary between hexamer columns, this produced 26 interface modes (13 polarisation doublets) 0.57–0.43 meV below E_{X}, 72–76 % photonic and with 87–93 % of their photon intensity on the 13 edge pillars,

i.e. the pillars inside the pump whose inter-cell partner lies outside (Fig. 2 of Report I). Lattices of expanded and shrunken hexamers are the photonic version of the Wu–Hu scheme [2], whose interfaces can host helical edge states, and an optically written interface would make such a channel reconfigurable. A direct test is to inject polaritons resonantly at one end of the interface and to look for propagation along it, with the incoherent pump as a gain medium that compensates the losses.

The present work simulates this experiment numerically. Section 2 introduces the time-dependent model, in which the blueshift is no longer prescribed but produced by an incoherent reservoir. Section 3 first establishes the steady state under the incoherent pump (Fig. 1), then shows resonant injection on the straight interface and on an interface truncated at a horizontal line (Figs. 2, 3), a frequency-domain transmission calculation (Fig. 4), and finally an analysis of the interface eigenmodes as an effective chain that explains the result quantitatively (Figs. 5, 6).

2 Time-dependent model

2.1 Equations

The fields are the same as in Report I: photons (E_x, E_y) in the linear basis and excitons (X_+, X_-) in the circular basis, with the identical linear Hamiltonian H_{lin} (photon potential of the etched lattice, photon and exciton kinetic energy, Rabi coupling), except that the static pump shift $U_X = B_{\mathrm{eff}}$ is removed. It is replaced by a real reservoir density $n_R(\mathbf{r},t)$ of incoherent excitons in the spirit of Ref. [8]:

$$i\hbar\,\partial_t\psi = [\,H_{lin} + V_X(n_X, n_R) - (i\hbar/2)(\Gamma - R\,n_R)\,]\,\psi + i\hbar F(\mathbf{r})e^{-iE_d t/\hbar} + noise,$$
$$\partial_t n_R = P(\mathbf{r}) - \gamma_R n_R - (R_{ex}|X|^2 + R_{ph}|E|^2)\,n_R.$$

The exciton potential V_X contains the saturable exciton–exciton interaction $g_1 n/(n_{\mathrm{sat}} + n)$ for the same spin and $g_2 = -0.1 g_1$ for the opposite spin, with $g_1 = g_{XX} n_{\mathrm{sat}}$ so that the low-density limit is g_{XX}, and the reservoir shift $g_R n_R$. Γ holds the photon and exciton loss rates γ_c, γ_x; the reservoir feeds both channels by stimulated scattering with rates R_{ph} and R_{ex}. Spontaneous scattering enters as white noise in the exciton channel with variance $R_{\mathrm{ex}} n_R \Delta t/(2\Delta A)$ per pixel; for the band structure of Fig. S2 the photon field additionally receives the corresponding noise of its gain channel, with variance $R_{\mathrm{ph}} n_R \Delta t/(2\Delta A)$ (Sec. 4). The inhomogeneous exciton line is a static exciton disorder potential, Gaussian with 1.5 meV FWHM, uncorrelated between pixels. The photon fields are normalised so that $|\psi|^2$ is a density in $\mu\mathrm{m}^{-2}$. F is the resonant cw drive, a Gaussian spot of 4 μm FWHM centred on the lowest edge pillar with a phase gradient $\exp(i k\,\hat{\mathbf{t}}\cdot\mathbf{r})$ along the edge direction $\hat{\mathbf{t}} = (\cos 60°, \sin 60°)$; its amplitude is chosen such that $|F|^2/(\gamma_c/2)^2$ equals a nominal injected density n.

2.2 Parameters

All lattice, cavity and exciton parameters are those of Report I (gpe_v5). The additional parameters are:

quantity	value	origin
photon loss $\hbar\gamma_c$	0.20 meV	= photon FWHM of Report I
homogeneous exciton loss $\hbar\gamma_x$	40 μeV	Ref. [4]

inhomogeneous exciton line	1.5 meV FWHM	Report I; static disorder, white on the 350 nm grid
exciton–exciton interaction g_{XX}	10 μeV μm²	Ref. [5]
opposite-spin interaction g_2/g_1	−0.1	Ref. [6]
saturation density n_{sat}	660 μm^{-2}	Ref. [7], (In,Ga)As
reservoir interaction g_R	$2g_{XX}$ = 20 μeV μm²	exchange factor 2 for a non-degenerate reservoir
reservoir decay $\hbar\gamma_R$	0.20 meV	Ref. [5]
photon gain R_{ph}	2.32×10^9 μm²/s	fixed by $g_R n_R^{th} = B_{eff}$ with $n_R^{th} = \gamma_c/R_{ph}$ = 131 μm^{-2}
exciton gain R_{ex}	0.1 R_{ph}	below γ_x/γ_c = 0.2, so that the lower polariton, not the exciton continuum, lases
pump unit P_{th}^{est}	$\gamma_R n_R^{th} = 3.97\times10^{13}$ $\mu m^{-2}s^{-1}$	linear threshold of a fully photonic mode
pump profile	P_0 / 0.1 P_0 / 0	pumped region / rest of the lattice / outside the lattice footprint (pillars + 1 μm); edges blurred with σ = 0.3 μm

The choice $R_{ph} = g_R\gamma_c/B_{eff}$ ties the model to the stationary design: at the photonic threshold the reservoir shifts the exciton by exactly the B_{eff} of Report I. With a linear gain the mode that reaches threshold first is the one with the smallest $(|C|^2\gamma_c + |X|^2\gamma_x)/(|C|^2R_{ph} + |X|^2R_{ex})$; for $R_{ex}/R_{ph} < \gamma_x/\gamma_c$ this is always the most photonic mode, for the opposite inequality the dark exciton continuum. No intermediate mode can win; energy relaxation is not part of the model (Sec. 4).

2.3 Numerics

The equations are integrated with a Crank–Nicolson scheme with predictor–corrector for the nonlinear and reservoir terms, time step 20 fs, on a 350 nm grid (124 × 126 points, the window of Report I) on the CPU, in a frame rotating at E_X − 1 meV. The grid spacing is a compromise for computational cost; as shown in Sec. 3.5, it introduces an on-site scatter of the edge-pillar energies that matters at the 10 μeV scale. Two pump geometries are considered: the straight 60° edge of Report I ("straight") and the same half-plane restricted to y < 32.1 μm, the hexamer-row boundary closest to the targeted truncation line ("cut"). Power-sweep points are first evolved until the photon number is stationary (at most 500 ps), then 1024 samples are recorded every 0.2 ps, which gives spectra per pixel with 20 μeV resolution. Injection runs use 100 ps of incoherent pumping, then 150 ps with the drive; the coherent response is obtained by demodulating the field at the drive frequency over the last 80 ps, which removes the incoherent emission of the reservoir. Edge profiles are this coherent intensity in Gaussian windows (σ = 0.6 μm) on the edge pillars, ordered along the edge and normalised to the injection pillar.

As a test of the time-dependent model against the eigenmode solver, the whole lattice is pumped uniformly at 0.3 P_{th}^{est}, well below threshold, and the photoluminescence far field is recorded in steady state (300 ps warm-up, 819 ps record sampled every 0.02 ps, 5 μeV resolution, slit k_y = 0, seven realisations with and four without exciton disorder). In these runs the spontaneous-scattering noise acts in both gain channels, i.e. the photon field receives the noise counterpart of its gain $R_{ph}n_R$ in addition to the exciton noise (variance $R_{ph}n_R\Delta t/(2\Delta A)$ per pixel and polarisation). The fine sampling is required because this noise also populates photon states tens of meV above the band, which a 0.2 ps sampling interval folds back into the

spectral window. The simulation is compared with the far field of the eigenmodes of the same lattice, computed with the time-averaged exciton shift of the simulation (0.75 meV), each mode weighted with its noise source (exciton and photon channel) divided by its net linewidth (Fig. S2). Without disorder the two band structures agree: at $k_x = 0$ the band bottom lies at −1.172 meV in the simulation and at −1.151 meV in the eigenmode spectrum, at $k_x = 1.4\ \mu m^{-1}$ at −0.566 and −0.586 meV, with comparable line widths (157 versus 167 μeV at $k_x = 0$). The exciton disorder broadens the lines to about 190–230 μeV and lowers the band by 85–140 μeV, but leaves the band shape intact; its emission background between 0 and 0.4 meV is 5 % of the band intensity.

3 Results

3.1 Steady state under the incoherent pump

Figure 1 characterises the system without drive. The photon number grows linearly with the pump up to 0.81 P_{th}^{est} and jumps by two orders of magnitude between 0.81 and 1.10 (Fig. 1b); undriven runs at 0.95–1.01 do not lase (Sec. 3.7), so the threshold lies between 1.01 and 1.10 P_{th}^{est}. The mean exciton shift in the pumped region rises from 0.13 meV at 0.05 P_{th}^{est} to 2.10 meV at 0.81 and saturates above threshold at 2.8–3.0 meV, where the reservoir is clamped near n_R^{th} (Fig. 1c); the weakly pumped rest of the lattice shifts ten times less. Below threshold the emission of the pumped region is a broad band dominated by the inhomogeneous exciton line, which moves up with the blueshift (Fig. 1d). Above threshold a single line at −0.78 meV (81 μeV FWHM) appears in the photon-like interior band of the pumped region, as expected for $R_{ex}/R_{ph} < \gamma_x/\gamma_c$; at 1.5 P_{th}^{est} the emission becomes multimode.

The interface of Report I is visible in the sub-threshold photoluminescence. At 0.81 P_{th}^{est} the photon spectrum on the edge pillars has a peak at −0.56 meV, inside the stationary interface band, where it exceeds the spectrum of all other pillars by a factor 14 (Fig. 1e; bulk-to-edge ratio 0.062, compared with 0.176 in Report I). The reservoir therefore produces the same interface as the prescribed B_{eff}, and the time-dependent model is a faithful extension of the stationary design.

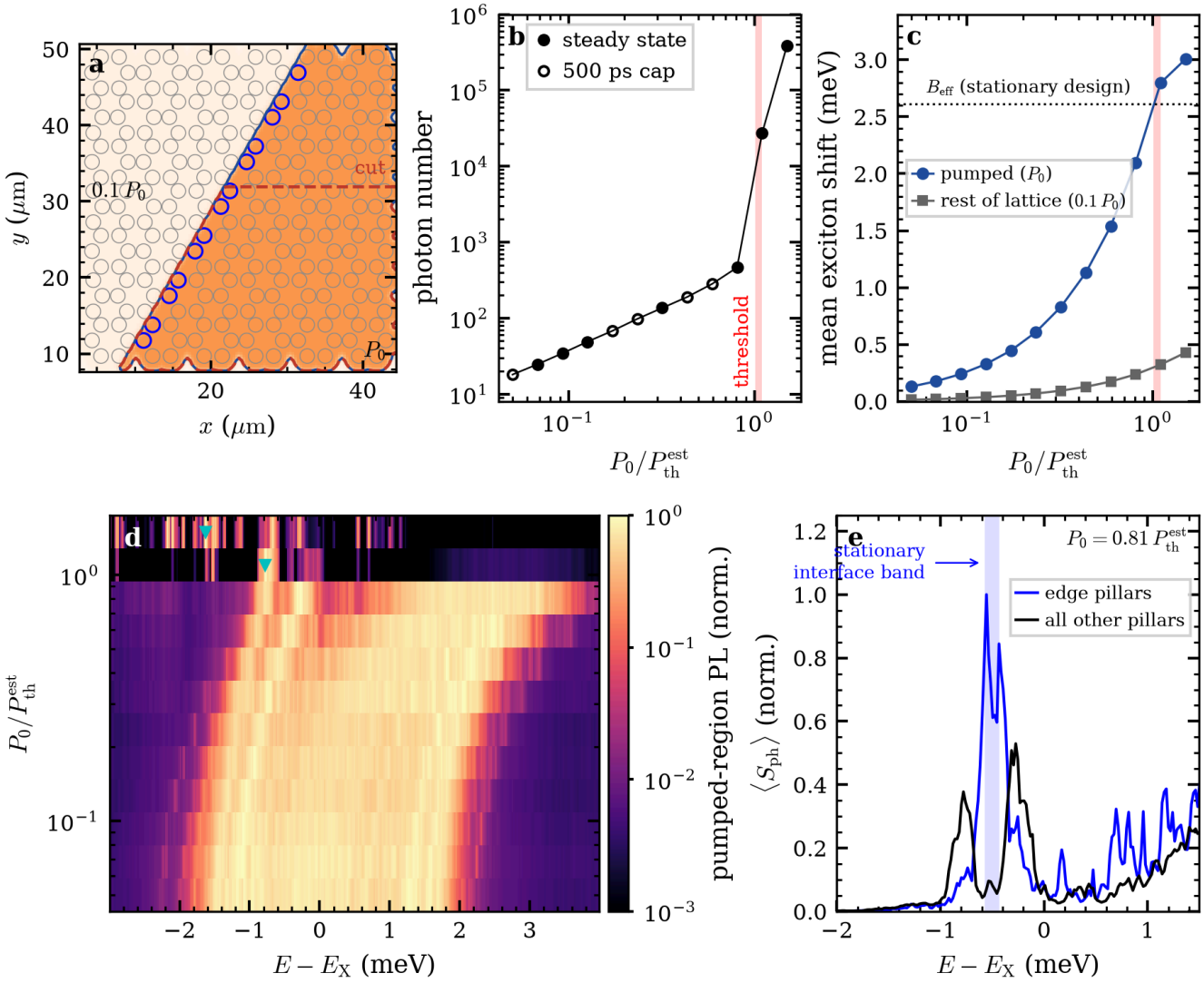


Figure 1. Steady state under a cw incoherent pump. a, Pump profile (orange: P_0 in the pumped half-plane, 0.1 P_0 on the rest of the lattice, zero outside), pillars (grey circles), the 13 edge pillars (blue), the straight pump edge (blue line) and the cut geometry (red dashed). **b**, Photon number versus pump power; open symbols: warm-up stopped at the 500 ps cap before the stationarity criterion was met. Red band: lasing threshold, between 1.01 (no lasing in undriven runs, Sec. 3.7) and 1.10. **c**, Mean exciton shift $g_{\rm R} n_{\rm R} + V_{\rm X}$ in the pumped region and on the rest of the lattice; dotted: $B_{\rm eff}$ of Report I. **d**, Photon spectra of the pumped region versus pump power, each normalised to its maximum (log scale); triangles: lasing lines. **e**, Mean photon spectrum on the edge pillars and on all other pillars at 0.81 $P_{\rm th}^{\rm est}$ (0.5 µm Gaussian windows, as the cost of Report I); shaded: interface band of the stationary design (150 nm grid). One disorder seed, 350 nm grid.

3.2 Resonant injection on two interfaces

The resonant drive is applied at the lowest edge pillar on top of an incoherent pump at 0.90 $P_{\rm th}^{\rm est}$, below threshold, so that no lasing occurs without the drive (confirmed by undriven controls at 0.85, 0.90 and 0.93). The drive is tuned into the interface band, $E_{\rm d} - E_{\rm X} = -0.58$ meV, with $k = +0.6$ µm^{-1} along the edge and a nominal density of 100 µm^{-2}. The animation in Fig. 2 shows the photon density after the drive is switched on, averaged over five disorder seeds for each geometry. The injected light fills the injection dimer and its hexamer within a few picoseconds and then stays there: the second dimer receives 2–9 % of the intensity of the injection dimer and the third another 0.1–1.5 % of that, after which the edge cannot be distinguished from the background emission of the reservoir (Fig. 2c, d). The space–time map along the edge (Fig. 2c) shows a stationary pattern from the first picoseconds on, with no front moving along the edge.

The profile along the edge is a staircase: it is nearly flat within each dimer and drops between dimers. The intensity ratio of consecutive dimers is 0.023–0.094 for the first and 0.001–0.015 for the second step (Fig. 2e), and the 1 % reach is 9.1 µm in all ten runs. The two geometries are indistinguishable up to the noise floor: for equal disorder seeds the profiles agree to the third digit, because the light never comes close to the corner at $s \approx 27$ µm where the cut interface turns. The coherent intensity stays on the edge (edge-to-bulk ratio 28–90 beyond 6 µm), but it does not travel. Without the exciton disorder (dotted) the decay is not slower but faster.

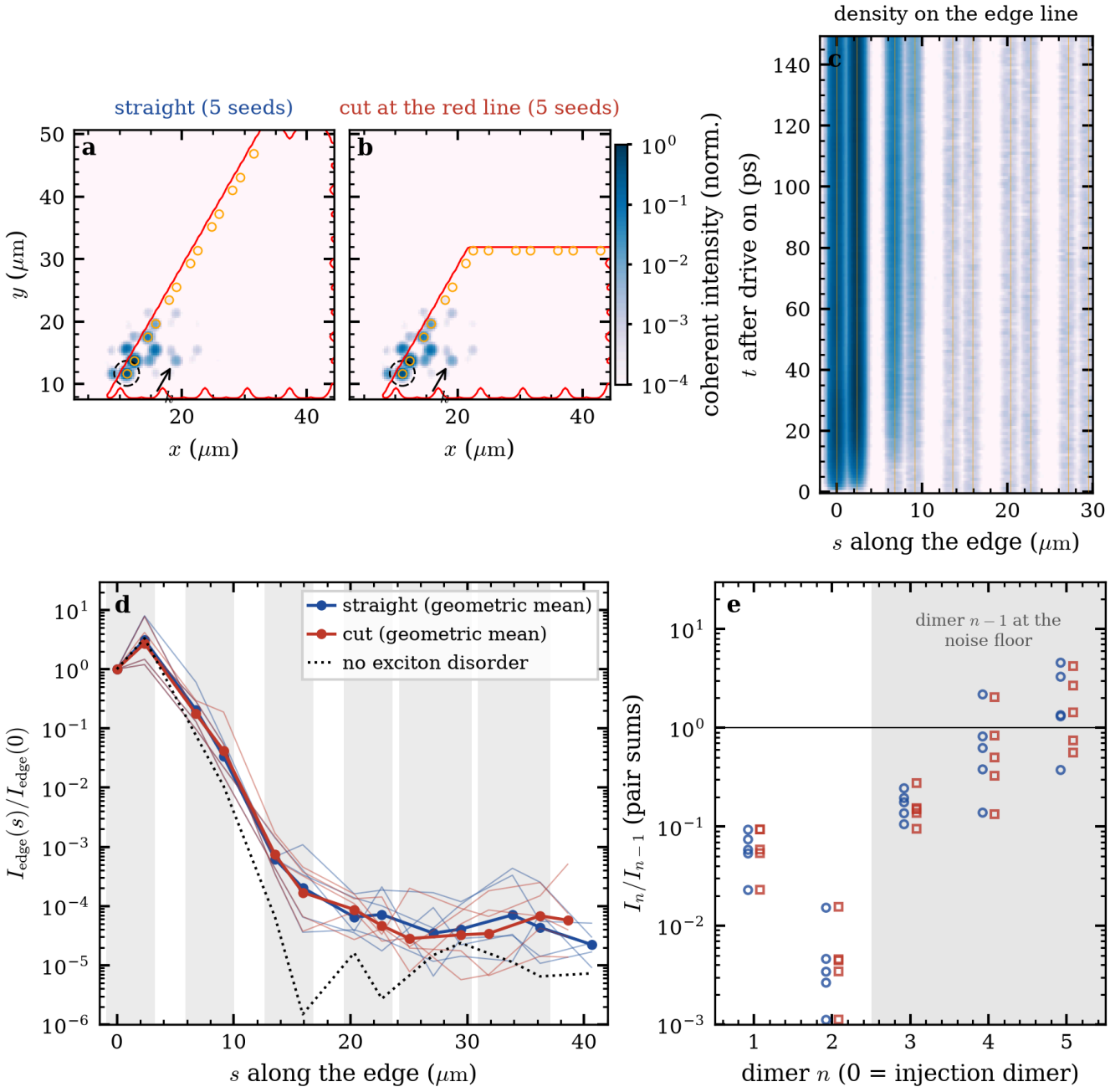


Figure 2. Resonantly injected light stays at the injection dimer on both interfaces. The HTML version of this report contains an animation of the photon density during the first 150 ps after the drive is switched on. **a, b**, Coherent intensity at the drive frequency (last 80 ps, seed mean, log scale) for the straight and the cut interface; dashed circle: drive spot (FWHM 4 µm), arrow: direction of k. **c**, Photon density (coherent and incoherent) along the edge line versus time after drive on, straight interface; orange lines: edge pillars. **d**, Coherent intensity on the edge pillars normalised to the injection pillar; thin lines: single seeds, symbols: geometric mean; dotted: straight interface without exciton disorder; grey bands: edge dimers. **e**, Intensity ratio of consecutive dimers (pair sums); grey: dimer $n-1$ already at the noise floor. Pump 0.90 P_{th}^{est}, $E_d - E_X = -0.58$ meV, $k = 0.6$ µm^{-1}, $n = 100$ µm^{-2}, E_x polarisation, 350 nm grid.

3.3 Independence of the drive

The staircase is not a matter of drive parameters (Fig. 3). Scanning the wave vector from −0.45 to +1.4 µm^{-1} changes the transfer into the second dimer between 0.0014 (backwards) and 0.065 (+0.6 to +1.0 µm^{-1}), a weak preference for the forward direction, but never the number of dimers that are reached (Fig. 3a). Neither does the drive energy across the interface band (−0.64 to −0.42 meV in 20 µeV steps, Fig. 3b), the pump power (0.85–0.93), the injected density (100 and 1000 µm^{-2}), the polarisation (E_x, E_y) or the drive energy at the band edges (Fig. 3c). Replacing the resonant drive by a small non-resonant spot (2 µm FWHM on the injection pillar, on top of the 0.9 P_{th}^{est} pump) gives the same picture (Fig. 3d, Fig. S1): a weak spot is lost in the reservoir emission, a spot of 2–8 P_{th}^{est} lases locally, and its emission falls to 10 % on the second edge pillar and to the background level within about 7 µm, spreading into the pumped bulk rather than along the edge.

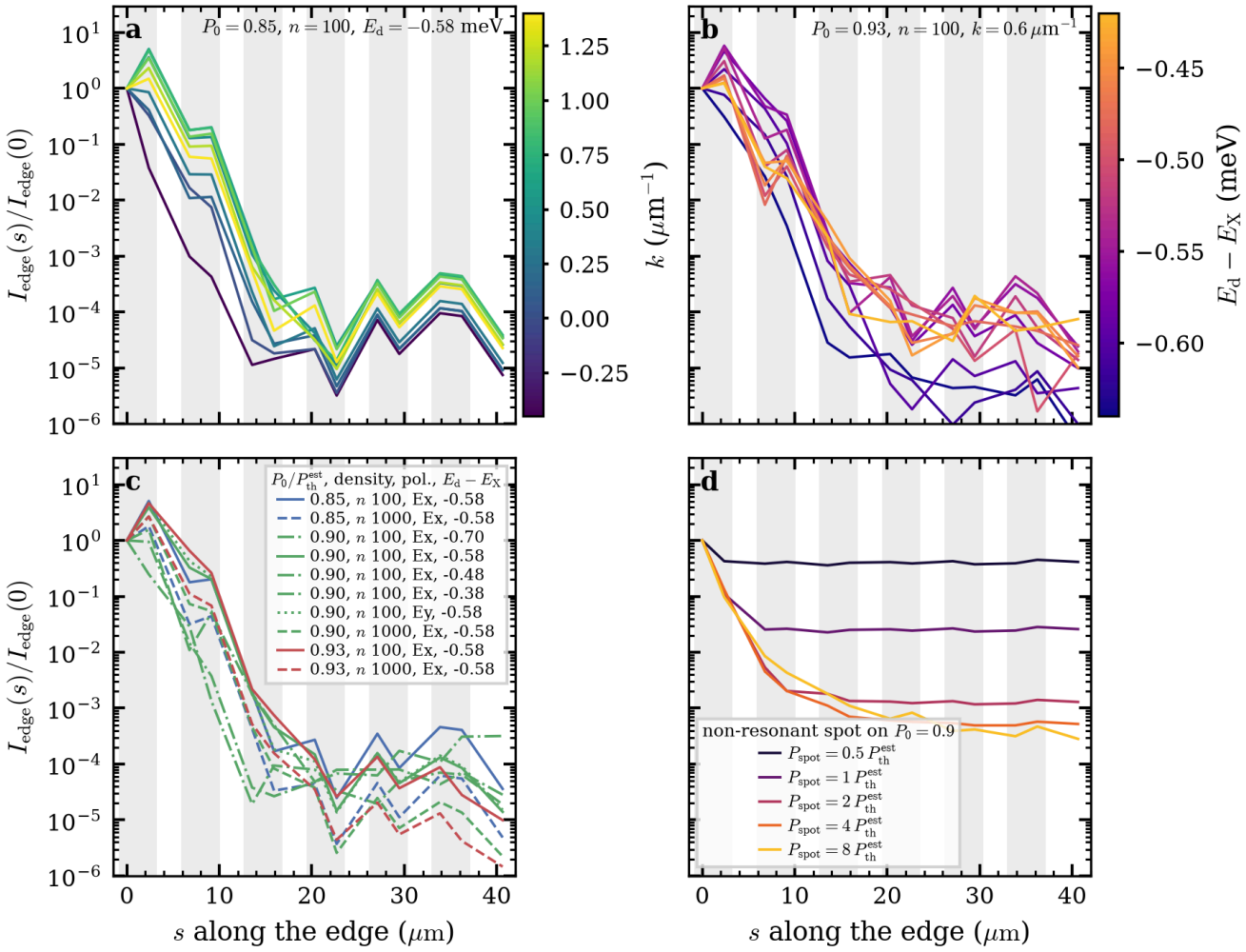


Figure 3. The decay along the edge does not depend on the drive. Coherent edge profiles as in Fig. 2d, straight interface, one seed each. **a**, Wave vector along the edge at 0.85 P_{th}^{est}. **b**, Drive energy across the interface band at 0.93 P_{th}^{est}. **c**, Pump power (colour), injected density (dashed: 1000 µm^{-2}), polarisation (dotted: E_y) and drive energy (dash-dotted). **d**, Non-resonant spot of 2 µm FWHM on the injection pillar instead of the resonant drive (time-averaged photon density, coherent and incoherent, normalised to the injection pillar). Grey bands: edge dimers.

3.4 Far field and transmission along the edge

The band structure that an experiment would record is the photoluminescence far field under the incoherent pump. It is obtained from long steady-state simulations with the pump of the straight interface at 0.90 $P_{\mathrm{th}}^{\mathrm{est}}$ (500 ps warm-up, 819 ps record, 5 μeV resolution; four realisations without and seven with exciton disorder), by Fourier transforming the photon field in time and along the edge direction ($k_{\parallel}$, slit $k_{\perp} = 0$), either over the whole lattice or after a real-space filter on the edge pillars (Gaussian windows of 1.2 μm), as with a spatial filter in the image plane of the experiment (Fig. 4a, b, d). The same procedure reproduces the eigenmode band structure of the uniformly pumped lattice (Fig. S2). The full far field (Fig. 4a) is dominated by the dispersive, photon-like interior band of the pumped region, from −0.85 meV at $k_{\parallel} = 0$ to −0.06 meV at $k_{\parallel} = 1.5\ \mu\mathrm{m}^{-1}$. The edge filter suppresses it and reveals the interface band (Fig. 4b): an emission line at −0.50 meV around $k_{\parallel} = 0$ and a second one at −0.31 to −0.37 meV at large $|k_{\parallel}|$. They are the bonding and the antibonding band of the edge chain of Sec. 3.5: the bonding band, in phase on the two pillars of a dimer (2.37 μm apart), radiates near $k_{\parallel} = 0$, the antibonding band near $\pi/(2.37\ \mu\mathrm{m}) \approx 1.3\ \mu\mathrm{m}^{-1}$; the gradual change of energy with $k_{\parallel}$ is this form factor, not a dispersion, since each band is only about 25 μeV wide. The far field of the eigenmodes, weighted like the simulated emission (exciton-channel source, gain-narrowed width per mode), agrees with the simulation to within 10–20 μeV (Fig. 4c). With exciton disorder the lines broaden and lie on a broad excitonic background (Fig. 4d, see Sec. 4), but the interface emission remains visible. The interface therefore carries a band that is bright in emission but, at the physical linewidths, non-dispersive; only with linewidths reduced 40-fold does the far field resolve the two cosine-shaped SSH bands (150 nm grid, not shown).

Transport is quantified by the response to a local drive in real space; a far-field transform of such a locally driven field is not a band structure, because the response is confined to about 1 μm around the source. To separate the properties of the interface from the nonlinear and stochastic parts of the model, the linearised equations are solved directly in the frequency domain. Below threshold and for a weak drive the field obeys

$$[\, H_{lin} + g_R n_R - E - (i\hbar/2)(\Gamma - R n_R)\,]\,\psi = i\hbar F, \quad n_R = P/\gamma_R,$$

which is solved for 241 energies with the same drive spot ($k = 0$) in four variants: the time-dependent model at 0.90 $P_{\mathrm{th}}^{\mathrm{est}}$ (with disorder and gain), the same without disorder, the same without gain, and the stationary design of Report I (static B_{eff}, no disorder, no gain), all on the 350 nm grid. In the stationary design the intensity along the edge falls by six orders of magnitude within 15 μm at every energy (Fig. 4e). The $1/e$ decay length, fitted between 5 and 30 μm, is 0.7–1.6 μm for the stationary design and 0.7–1.9 μm without gain at all energies; gain lengthens it to a median of 1.5–2.0 μm between −0.7 and −0.3 meV and to at most 5.7 μm at other energies; removing the disorder changes little (Fig. 4f). The propagation is therefore limited by the interface of the design itself, not by the time-dependent model.

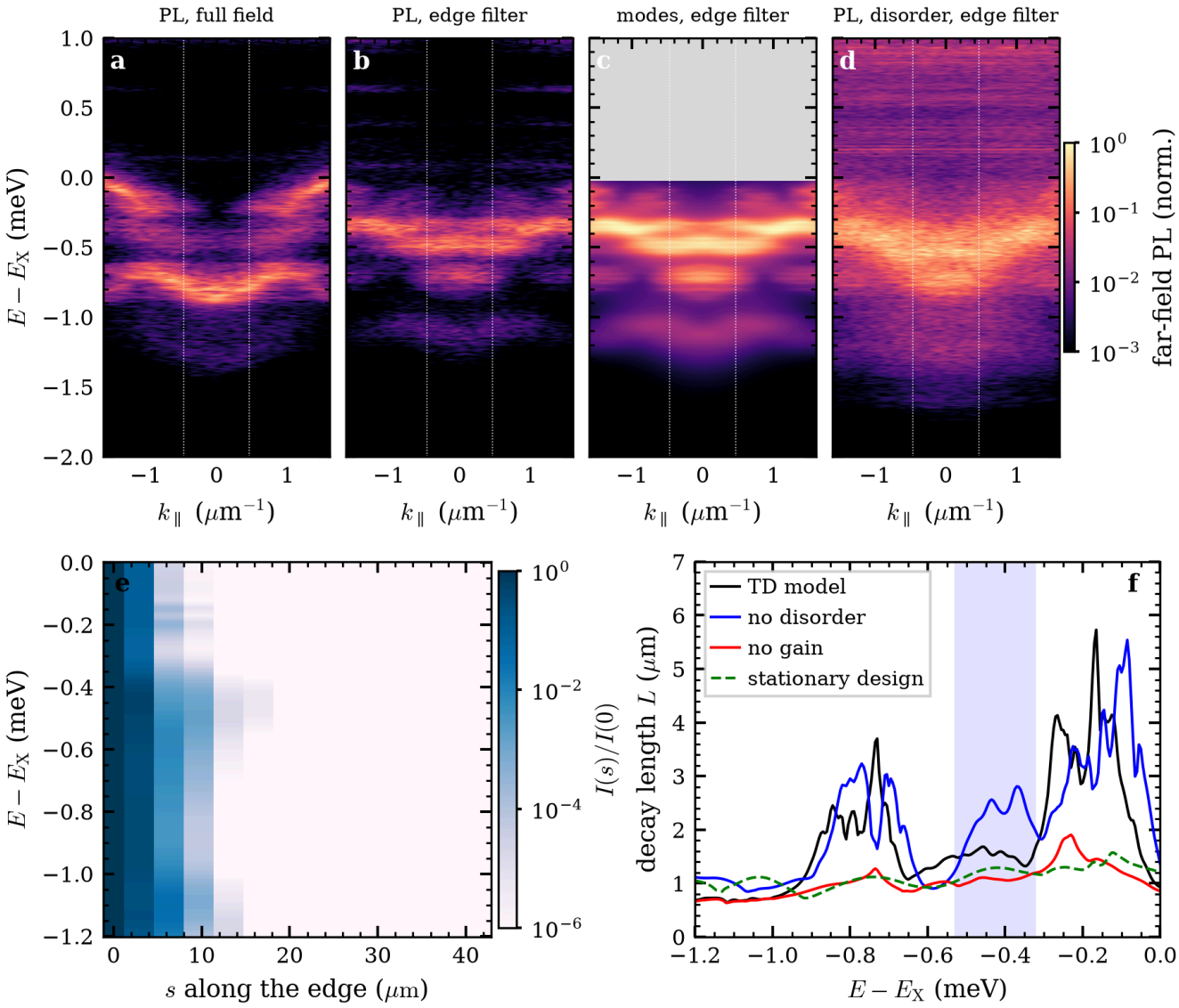


Figure 4. In emission the interface band is bright but flat, and a local drive does not propagate. a–d, Photoluminescence far field along the edge direction ($k_\perp$ = 0) under the incoherent pump of the straight interface at 0.90 P_{th}^{est}, 350 nm grid. **a**, Simulation, whole lattice; **b**, simulation with a real-space filter on the 13 edge pillars (Gaussian windows, σ = 1.2 µm); both without exciton disorder, mean of four realisations, 819 ps record, Hann window. **c**, Eigenmodes of the same lattice with the time-averaged exciton shift of the simulation, edge filter, each mode weighted with its exciton-channel noise source and broadened by its gain-narrowed linewidth (grey: above the computed spectral window). **d**, As b with 1.5 meV exciton disorder, mean of seven realisations. Dotted: $\pm\pi/a_{latt}$; log scale, each panel normalised. **e**, Intensity on the edge pillars relative to the injection pillar versus energy for a local drive (4 µm FWHM, E_x, k = 0, lowest edge pillar), stationary design. **f**, $1/e$ decay length of the edge intensity (fit 5–30 µm) for the four variants of the linear response (TD model: disorder and gain at 0.90 P_{th}^{est}); shaded: interface band on the 350 nm grid.

3.5 The interface as an SSH chain

The nature of the interface band follows from the eigenmodes of the stationary design. Since TE–TM and Zeeman splitting vanish, the x-polarised sector (E_x with the exciton combination $(X_+ + X_-)/\sqrt{2}$) and the y-polarised sector are exact copies; the analysis is therefore restricted to the 13 interface modes of the x sector, computed on the 150 nm grid of Report I (shift-invert solve, 160 modes around −0.52 meV). The photon amplitude a_j of each mode on pillar j is obtained by projecting E_x on a normalised Gaussian (σ = 0.5 µm) at the pillar centre. The 13 interface modes carry 88–95 % of $\Sigma|a_j|^2$ on the 13 edge pillars, and

their edge amplitudes form an almost orthonormal set (singular values 0.99–1.03). Orthonormalising them (Löwdin) gives an effective Hamiltonian H_{eff} on the edge pillars that reproduces the 13 energies exactly.

The edge pillars lie on a single straight line and alternate between two distances (Fig. 5a): 2.369 μm (= $a + s$, the two edge pillars of the same boundary hexamer, 0.45 μm apart edge to edge) and 4.404 μm (= $2a - s$, to the next hexamer, with a 2.49 μm gap and no overlap). H_{eff} is a textbook Su–Schrieffer–Heeger (SSH) chain [3] (Fig. 5c): uniform on-site energies (−0.500 meV, scatter ±4 μeV), an intra-dimer hopping t_1 = −59.5 μeV across the short bond, an inter-dimer hopping t_2 = +12.0 μeV across the long one, $|t_2/t_1|$ = 0.20, and next-nearest couplings below 0.6 μeV. A chain of six identical two-site cells reproduces the 12 paired mode energies to about 1 μeV; the 13th mode, at −0.5008 meV, is the end state of the unpaired last pillar (Fig. 5b). The modes are standing waves of the chain: their dominant wave vector along the edge places each of them on the Bloch bands of the infinite chain (Fig. 5d). These are a bonding band (−0.571 to −0.550 meV, 22 μeV wide, maximum at k = 0) and an antibonding band (−0.455 to −0.429 meV, 26 μeV wide), separated by a 94 μeV gap. In tight-binding terms each boundary hexamer contributes one dimer of dangling pillars, whose inter-cell partners were removed by the pump edge; neighbouring dimers couple only indirectly through the bulk, hence the small t_2.

A band of width $2|t_2|$ has a maximum group velocity $a_{\mathrm{latt}}|t_2|/\hbar$; here $|v_g| \leq 0.135$ μm/ps (Fig. 5e). With the polariton linewidth $\Gamma = |C|^2\hbar\gamma_c + |X|^2\hbar\gamma_x$ = 158 μeV ($|C|^2$ = 0.74), i.e. a lifetime of 4.2 ps, the intensity decays along the edge within $L = v_g\tau \leq 0.56$ μm. The linewidth exceeds the band width sevenfold, which is why the bands appear as flat lines in the far field (Fig. 4b, c).

On the 350 nm grid of the time-dependent runs the hoppings are almost the same (t_1 = −56 μeV, t_2 = +13.5 μeV), but the on-site energies scatter between −80 and +70 μeV (Fig. 5f): the coarse pixelation of the pillars acts as an on-site disorder much larger than t_2 and localises the interface modes on one or two dimers. This grid artefact comes on top of the exciton disorder; it does not change the conclusion, because the propagation length is below a micrometre even on the clean grid, but the 350 nm grid is not adequate for any interface whose couplings are at the 10 μeV scale.

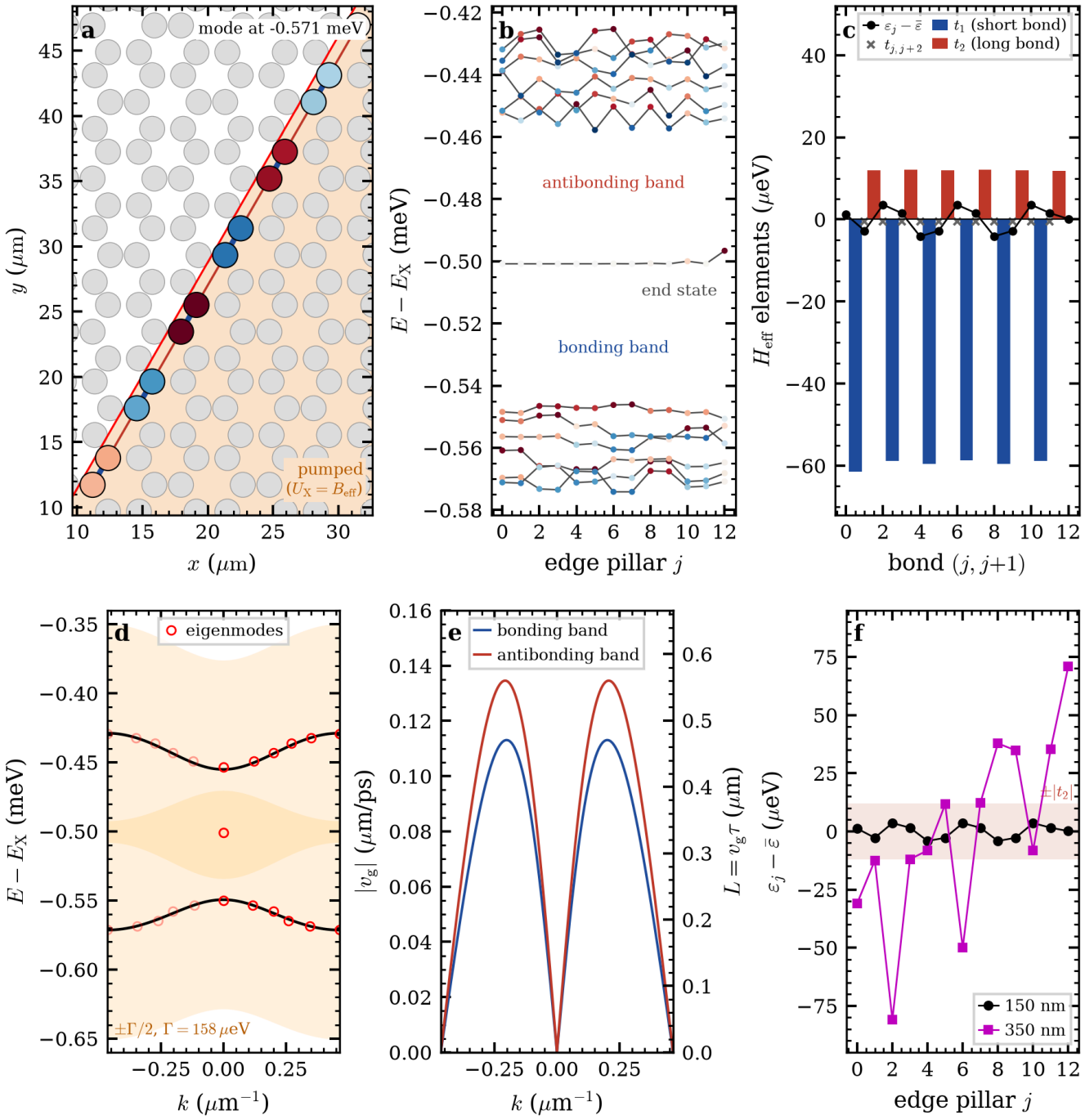


Figure 5. The interface band is an SSH chain of weakly coupled edge dimers. Stationary design of Report I, x-polarised sector, 150 nm grid. **a**, Edge pillars (coloured by the amplitude of the lowest interface mode, red/blue = sign) on the pump edge (red line; orange: pumped side); blue thick lines: short bonds (t_1), red thin lines: long bonds (t_2). **b**, Amplitudes a_j of all 13 interface modes on the edge pillars, drawn at the mode energy (colour and vertical offset proportional to a_j). **c**, Elements of the effective Hamiltonian: nearest-neighbour hoppings across short (blue) and long (red) bonds, on-site energies relative to their mean (black), next-nearest couplings (crosses). **d**, Bloch bands of the periodic two-site chain (black) with $\pm\Gamma/2$ (shaded) and the interface modes at their dominant standing-wave k (red; faint: mirrored). **e**, Group velocity of both bands and propagation length $L = v_g\tau$ (right axis, $\tau = \hbar/\Gamma = 4.2$ ps). **f**, On-site energies of H_{eff} on the 150 nm and on the 350 nm grid; shaded: $\pm|t_2|$.

3.6 Chain model versus simulation

The chain explains the time-dependent result without free parameters. Gain narrows the line: below threshold the reservoir density is $n_R = p\, n_R^{th}$ at a pump p in units of P_{th}^{est}, so the net linewidth of an

interface mode is

$$\Gamma_{net}(p) = |C|^2\hbar\gamma_c(1-p) + |X|^2\hbar(\gamma_x - 0.1\,p\,\gamma_c),$$

21 μeV at $p = 0.9$ instead of 158 μeV without pump. Driving the 13-site H_{eff} with the same 4 μm spot and $k = 0.6$ μm^{-1}, $\psi = (E + i\Gamma/2 - H_{\mathrm{eff}})^{-1}F$, gives the profiles in Fig. 6a (band: 10–90 % range over drive energies across the interface band). With Γ_{net} the chain reproduces the size of the first two steps of the simulated staircase, and all simulated points down to the noise floor lie within its range; without gain it decays much faster than the simulation. The fitted decay lengths of the simulation, 3.3–5.5 μm, match the chain estimate $L_{\max} = a_{\mathrm{latt}}|t_2|/\Gamma_{\mathrm{net}} = 4.0$ μm.

> **Requirement for routing.** In a chain of dimers coupled by t_2 the propagation length is at most $L_{\max} = a_{\mathrm{latt}}|t_2|/\Gamma$. To cross the 40 μm interface, $|t_2|/\Gamma$ must reach about 6 (Fig. 6b): with the present $|t_2| = 12$ μeV a net linewidth of 2 μeV, or, at the present linewidths, an inter-dimer coupling of about 1 meV (no gain) or 0.13 meV (gain at 0.9 $P_{\mathrm{th}}^{\mathrm{est}}$), i.e. 10–80 times the present value. Gain alone cannot close the gap: once Γ_{net} falls below the energy mismatch of neighbouring dimers, $|t_2|$ must also exceed that mismatch (Sec. 3.7).

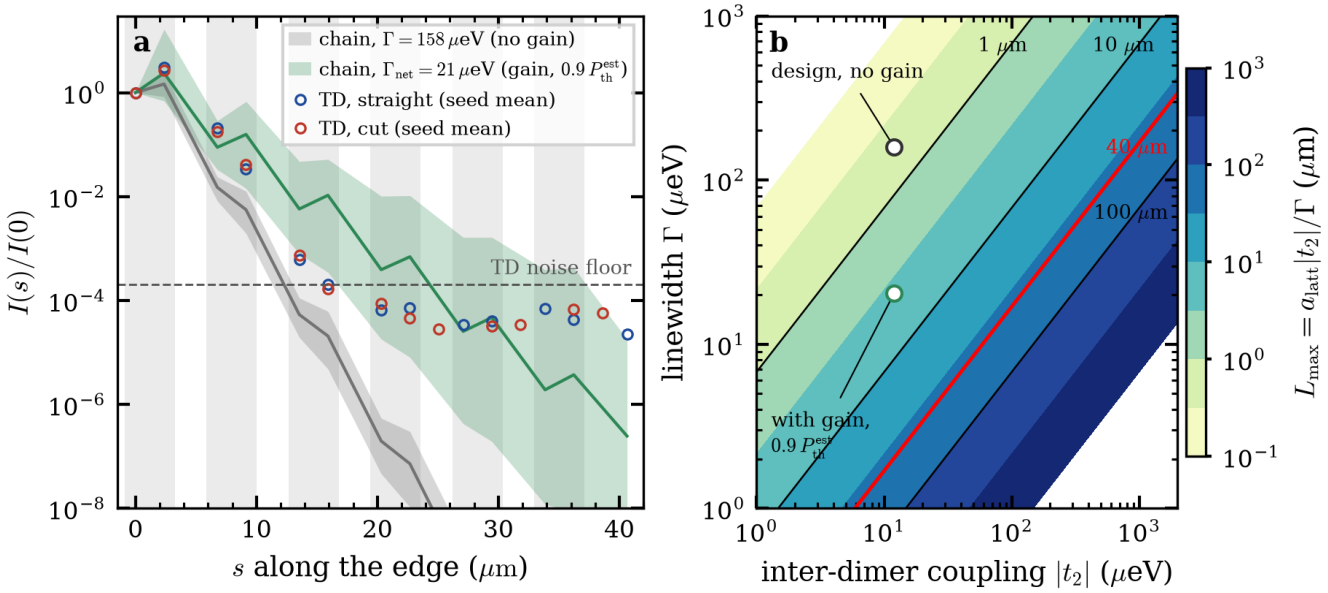


Figure 6. The chain model reproduces the simulated decay and sets the requirement for routing. a, Edge profiles of the time-dependent simulation (seed means of Fig. 2d, symbols) and of the 13-site chain H_{eff} of Fig. 5 driven like the simulation, with the bare linewidth (grey) and with the net linewidth at 0.9 $P_{\mathrm{th}}^{\mathrm{est}}$ (green); lines: geometric mean over drive energies across the interface band ±30 μeV, bands: 10–90 % range; dashed: noise floor of the simulation. **b**, Maximum propagation length $a_{\mathrm{latt}}|t_2|/\Gamma$ versus inter-dimer coupling and linewidth; red: 40 μm, the length of the interface; circles: present design without gain and with gain at 0.9 $P_{\mathrm{th}}^{\mathrm{est}}$.

3.7 Pump near threshold, strong injection

Two further parameters could in principle extend the reach: a larger gain, with the pump closer to threshold, and a stronger drive. Both are examined on the straight interface for one disorder seed, with all other parameters as in Fig. 2 (Fig. 7). Each pump value is complemented by an undriven control. None of

the controls at 0.95, 0.97, 0.99 and 1.01 $P_{\mathrm{th}}^{\mathrm{est}}$ lases within the 250 ps of the run: the reservoir in the pumped region stays at 0.990–0.992 of its undepleted value $p\,n_{\mathrm{R}}^{\mathrm{th}}$, compared with 0.957 in the lasing state at 1.10. The threshold therefore lies between 1.01 and 1.10 $P_{\mathrm{th}}^{\mathrm{est}}$, consistent with the linear estimate for the photon-like interior band ($|C|^2 \approx 0.87$), 1.015; the interface modes themselves would reach $\Gamma_{\mathrm{net}} = 0$ only at 1.034. A slow build-up beyond 250 ps at 1.01 cannot be excluded.

Despite the reduction of Γ_{net} from 21 μeV at 0.90 to about 4 μeV at 1.01, the profiles do not change (Fig. 7a). The transfer into the next dimer stays at 5–7 % and the one after at about 1 %, and both even decrease slightly towards threshold (Fig. 7c). The chain model accounts for the saturation of the gain effect: as soon as Γ_{net} drops below the energy mismatch between neighbouring dimers, the transfer is limited by that mismatch rather than by the loss. On the 350 nm grid the pixelation already provides on-site differences of tens of μeV (Fig. 5f), and the chain with these on-site energies predicts an almost flat transfer of 3.5–6 % per step; the clean 150 nm chain still predicts 7.5–11 %. The first step of the simulation lies between the two predictions. The second step is 3–5 times lower than even the 350 nm chain, and the same holds without exciton disorder (Fig. 2d, dotted). The discrepancy is not caused by the reservoir-written interface: the same eigenmode analysis with the exciton shift $g_{\mathrm{R}} n_{\mathrm{R}}$ of the time-dependent model at 0.90 $P_{\mathrm{th}}^{\mathrm{est}}$ (2.35 meV in the pumped region, 0.1 P_0 elsewhere) gives the same chain, $t_1 = -57$ μeV, $t_2 = +12.2$ μeV, with the same on-site pattern. The origin of the remaining factor of 3–5 remains open; the chain model with uniform gain constitutes an upper bound for the transfer in the simulation.

Raising the injected density from 100 to 1000, 3000, 10^4 and 3×10^4 μm^{-2} (nominal) does not extend the reach either (Fig. 7b). The density on the injection pillar grows sublinearly, from 36 to 1230 μm^{-2}, and above about 1000 μm^{-2} it exceeds the saturation density $n_{\mathrm{sat}} = 660$ μm^{-2}, beyond which the interaction model is no longer quantitative. The first transfer step decreases from 5.4 % to 0.6–2.7 %, as expected when the interaction blueshift detunes the injection dimer from its neighbour. The profiles fall to the noise floor at the same dimer; the floor itself lies lower only because the profiles are normalised to a larger injected intensity.

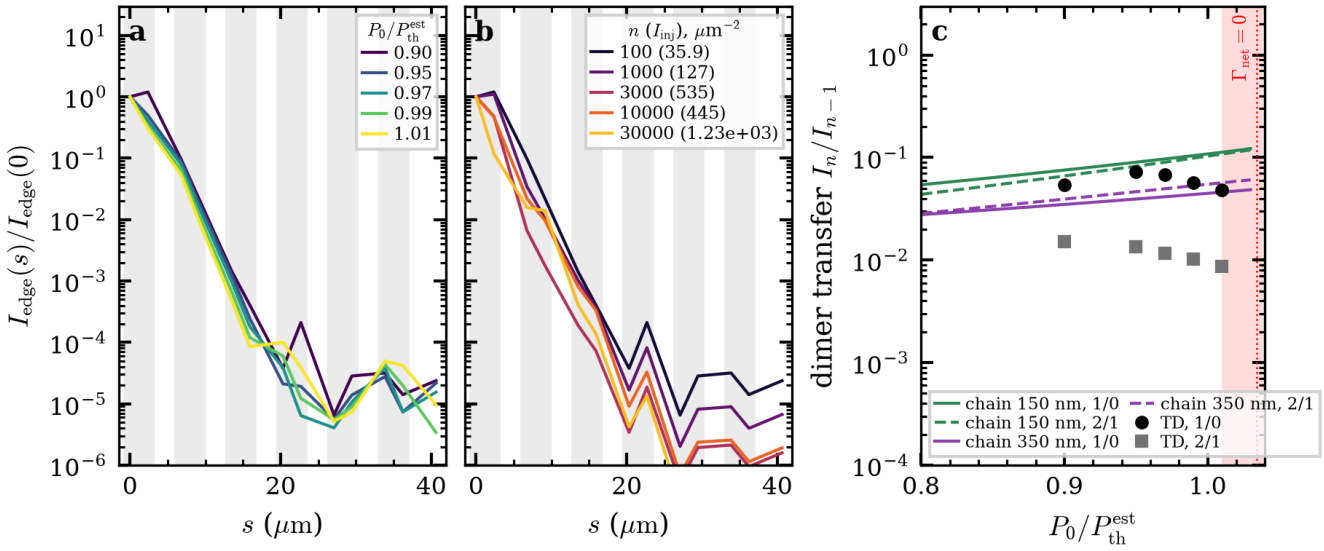


Figure 7. Neither a pump at threshold nor a strong drive extends the reach. a, Coherent edge profiles for incoherent pumps from 0.90 to 1.01 $P_{\rm th}^{\rm est}$ (none of the undriven controls lases). **b**, Profiles for nominal injected densities from 100 to 3×10^4 µm^{-2} at 0.90 $P_{\rm th}^{\rm est}$; in brackets: coherent density on the injection pillar. **c**, Intensity ratio of consecutive dimers (pair sums) of the simulation (symbols) and of the chain model with $\Gamma_{\rm net}(p)$, for the clean 150 nm chain (green) and the 350 nm chain with its pixel-induced on-site scatter (purple); geometric means over drive energies across the interface band. Red dotted: $\Gamma_{\rm net}$ = 0 for the interface modes; shaded: lasing threshold of the simulation (between 1.01 and 1.10). Straight interface, seed 1, $E_{\rm d} - E_{\rm X}$ = −0.58 meV, k = 0.6 µm^{-1}, 350 nm grid.

3.8 Bloch bands of the interface

The interface band of the finite structure (13 edge pillars) is confirmed by a Bloch calculation that removes all finite-size and termination effects. The supercell is a strip of the lattice one lattice constant wide and ten hexamer rows high (6.77 × 58.66 µm^2, 60 pillars), Bloch-periodic in both directions (Fig. 8a). The interface runs along the lattice vector $\mathbf{a}_1$ between two hexamer rows and cuts the two inter-cell bonds per hexamer that connect neighbouring rows; by the sixfold symmetry of the lattice this is the same cut as the 60° pump edge of Report I. With the bottom five rows blueshifted, the periodic cell contains two equivalent interfaces (in its centre and at its boundary) and no lattice termination. Parameters and grid (150 nm) are those of Report I. At 31 wave vectors $k_{\rm x}$ along the interface the full s band (120 modes, two per pillar) is computed. Each band is coloured by its interface localisation $\langle P_{\rm IF}\rangle = \Sigma_{\rm edge}\, w_j|\mathbf{E}|^2 \,/\, \Sigma_{\rm all}\, w_j|\mathbf{E}|^2$, the photon weight on the four edge pillars of the cell (pumped-side partners of the cut bonds, the definition of Report I) relative to all pillars, and drawn with an opacity given by its far-field intensity at $k_\parallel = k_{\rm x} + G$ (slit $k_{\rm y}$ = 0, extended zone).

Without blueshift the s band spans −1.49 to −0.98 meV in two groups separated by about 0.26 meV (Fig. 8b); with the blueshift everywhere it moves to −0.91 to −0.02 meV, the more excitonic upper group by more (Fig. 8c). In both cases $\langle P_{\rm IF}\rangle$ stays at or below 0.25 (mean 0.067, the share of edge pillars). With the interface (Fig. 8d) both sets of bulk bands appear, and inside the bulk gap between −0.715 and −0.294 meV two additional bands with $\langle P_{\rm IF}\rangle$ = 0.87–0.92 and $|C|^2$ = 0.70–0.76 emerge: a lower one from −0.581 to −0.553 meV (maximum at $k_{\rm x}$ = 0) and an upper one from −0.464 to −0.431 meV (minimum at $k_{\rm x}$ = 0), each four-fold per k (two polarisations, two interfaces). They are the bonding and antibonding bands of the edge chain of Sec. 3.5, now as Bloch bands of an infinite interface: positions, the sign of the curvature at $k_{\rm x}$ = 0 and the widths (28 and 33 µeV, compared with 22 and 26 µeV for the 13-site chain) agree. The Bloch calculation thereby confirms that the interface carries a genuine but extremely flat band, isolated from the

bulk by more than 0.13 meV on either side. In the far field of the whole cell the interface bands reach about 6 % of the brightest bulk band, because the interface occupies only a small part of the cell; spatial filtering on the interface (Fig. 4b) is required to see them in an experiment.

In each panel the 120 lowest eigenmodes per wave vector are computed, which in b and c is exactly the *s* band. In d this is no longer the case: above −0.27 meV the unpumped half contributes a dense continuum of exciton-like states ($|C|^2$ = 0.04–0.05, about 2000 states per meV), mostly excitons under the etched barrier between the pillars, which are pushed slightly below E_X by their coupling to the barrier photon states. These nearly dark states occupy the last of the 120 places, so the computed spectrum of d ends at −0.256 meV, and the upper bands of the pumped half (−0.25 to −0.02 meV in c) are not resolved in d (grey band in Fig. 8d). The same continuum exists in b and c but lies above the *s* band (b) or 2.6 meV higher (c). Resolving all states up to E_X would require several thousand modes per wave vector; the interface bands and everything below −0.256 meV are complete.

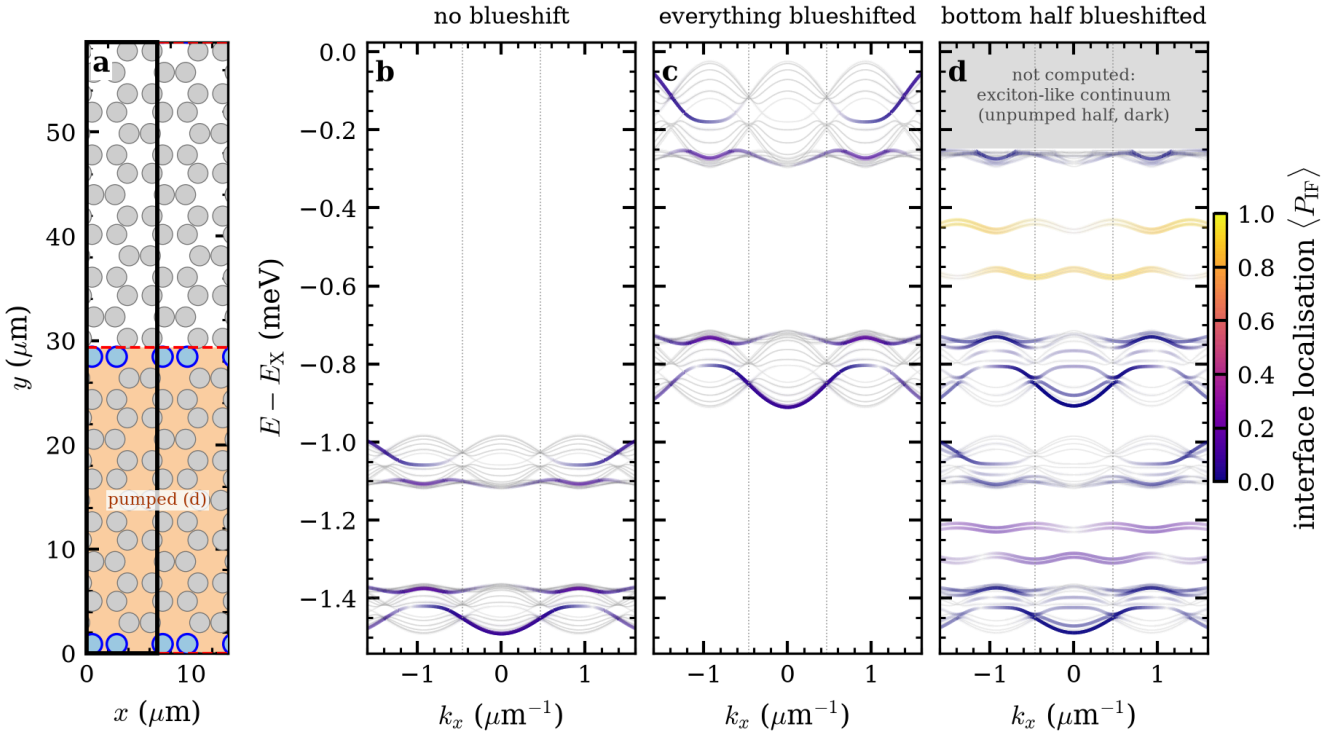


Figure 8. Bloch bands of a supercell across the interface. a, Simulation cell: the supercell (black frame, one lattice constant wide and 10 hexamer rows high, 60 pillars) is Bloch-periodic in x and y; the panel shows it together with its periodic images (pillars in grey). Orange: blueshifted region of d; red dashed: interfaces; blue: edge pillars used for $\langle P_{IF} \rangle$. **b–d**, The 120 lowest Bloch modes (2 per pillar) versus k_x along the interface without blueshift (b), with B_{eff} = 2.61 meV everywhere (c) and in the bottom half only (d, blurred edge σ = 0.3 µm). In b and c these form the complete *s* band; in d the dense, nearly dark exciton-like continuum of the unpumped half takes the highest places above −0.27 meV, so the upper bands of the pumped half are not computed (grey band, see text). Colour: interface localisation $\langle P_{IF} \rangle$; opacity: square root of the far-field intensity at $k_x + G$ (slit k_y = 0) relative to the brightest band of the panel; faint grey: all bands. The bands are shown in the extended zone, dotted: $\pm\pi/a_{latt}$. Parameters of Report I, 150 nm grid, 31 k-points in $[0, \pi/a_{latt}]$ mirrored by symmetry.

4 Discussion and limitations

- **Nature of the interface.** The pump changes only the exciton energy, not the ratio of intra- to inter-cell coupling: both sides of the interface are the same expanded-hexamer lattice, one of them detuned. The interface modes are therefore the states of the dangling pillars of the cut inter-cell bonds, an SSH-like chain whose inter-dimer coupling is mediated by the bulk, not a band-inversion

interface between two phases of the Wu–Hu scheme. No topological invariant is computed here; the conclusion on transport rests on the chain parameters, not on topology.

- **Grid.** The time-dependent runs use 350 nm. There the hoppings agree with the 150 nm values to about 10 % and the interface band lies up to 0.1 meV higher, but the pixelation adds an on-site scatter of ±70 μeV on the edge pillars (Sec. 3.5). The conclusion is grid independent because the clean 150 nm chain is already non-propagating; quantitative transport studies of improved designs need ≤ 150 nm.
- **Energy relaxation.** The model has no relaxation within the polariton band. It lases in the most photonic mode (the pumped interior band), not at the band top as observed on the KPZ samples; the threshold is the threshold of that mode. This affects the threshold value and the competition between modes, not the linear transport along the edge.
- **Exciton disorder.** The inhomogeneous exciton line is modelled as a static Gaussian exciton potential, uncorrelated between pixels, with RMS 1.5 meV/2.355. In a planar test cavity this reproduces the bare exciton absorption line of 1.5 meV FWHM on grids from 350 to 75 nm (1.49–1.52 meV) and for a correlated potential with l_c = 100 nm (1.50 meV); only white noise on a 50 nm grid narrows it (1.34 meV), because the exciton then moves between pixels. The lower polariton at $|X|^2$ = 0.28 is broadened by only 0–25 μeV and redshifted by 60–100 μeV, almost independently of the grid: motional narrowing makes the polariton inhomogeneous width far smaller than the $|X|^2 \times 1.5$ meV estimate used for the spectra of Report I.
- **Noise.** With spontaneous-scattering noise only in the exciton field, the photon-like polaritons are fed only through their exciton fraction, while localised, nearly dark excitons are fed directly and, without energy relaxation, decay only with γ_X. Their emission then forms a broad background of about half the band intensity (between 0 and 0.4 meV with disorder). Adding the noise counterpart of the photon gain lowers this background ninefold, to 5 % (Fig. S2). Figure S2 uses the complete noise model; the power sweep (Fig. 1), the photoluminescence far fields of Fig. 4a–d and the injection runs use exciton-only noise, sampled every 0.2 ps. For the coherent injection results the noise model only sets the noise floor; in Figs. 1 and 4 the excitonic background (most visible in Fig. 4d) is therefore overestimated by about an order of magnitude.
- **Linewidths.** The photon loss (0.2 meV) is the stationary linewidth of Report I; a higher-Q sample lowers Γ and lengthens L_{max} proportionally (Fig. 6b).
- **Quantitative gap.** The chain model with a uniform net linewidth brackets the first transfer step of the simulation, but overestimates the second by a factor 3–5, with and without exciton disorder (Sec. 3.7). Candidates are the spatially non-uniform gain near the blurred pump edge and losses into bulk modes through the 4 μm drive spot; they would lower, not raise, the transfer, so the chain remains an upper bound.
- **Drive density.** Above about 1000 μm^{-2} on the injection pillar the injected density exceeds n_{sat}; strong-coupling saturation is then only represented by the saturable interaction, not by a reduction of the Rabi splitting.
- **Statistics.** The main comparison is based on ten disorder realisations (five per geometry). All of them show the same staircase and the same 9.1 μm reach, so a larger ensemble would not change the conclusion; the realisation-to-realisation spread of the dimer ratios is shown in Fig. 2e. The scans of Figs. 3 and 7 use a single realisation per parameter value.
- **Convergence in time.** Five of the twelve power-sweep points (0.05, 0.17, 0.23, 0.44, 0.59) stopped at the 500 ps warm-up cap before the stationarity criterion was met; the points that define the threshold and the interface contrast (0.81, 1.10) are converged.

5 Conclusion

The time-dependent model confirms the interface of Report I: below threshold the reservoir writes the same interface band, bright on the edge pillars in photoluminescence. It does not support transport. Resonantly injected light stays at the injection hexamer on both interface geometries and for all drive parameters, and neither gain from an incoherent pump up to the lasing threshold, nor a strong drive, nor a non-resonant spot changes that. The reason lies in the stationary design: the interface band is an SSH chain of edge dimers with an inter-dimer coupling of 12 μeV, seven times narrower than the polariton line, so light decays within one or two dimer spacings; near threshold, where gain nearly cancels the loss, the transfer becomes limited by the energy mismatch of neighbouring dimers instead. The chain parameters bound the simulated decay from above and reproduce its first step. A routing experiment along an optically written interface therefore needs a design in which consecutive edge dimers are coupled directly, by about two orders of magnitude more than now, or a sample with a correspondingly narrower polariton line; the edge-chain analysis of Sec. 3.5 is a fast criterion for such a design search.

References

1. Report I: Optically written interface states at the pump boundary of a stretched-hexamer exciton-polariton lattice: a numerical study, working report, 9 October 2026 (numerical_report.html).
2. L.-H. Wu and X. Hu, Phys. Rev. Lett. **114**, 223901 (2015).
3. W. P. Su, J. R. Schrieffer and A. J. Heeger, Phys. Rev. Lett. **42**, 1698 (1979).
4. A. Delteil *et al.*, Nat. Mater. **18**, 219 (2019).
5. S. Harder *et al.*, ACS Photonics **8**, 1377 (2021).
6. M. Vladimirova *et al.*, Phys. Rev. B **82**, 075301 (2010).
7. D. Porras *et al.*, Phys. Rev. B **66**, 085304 (2002).
8. M. Wouters and I. Carusotto, Phys. Rev. Lett. **99**, 140402 (2007).

Supplementary figures

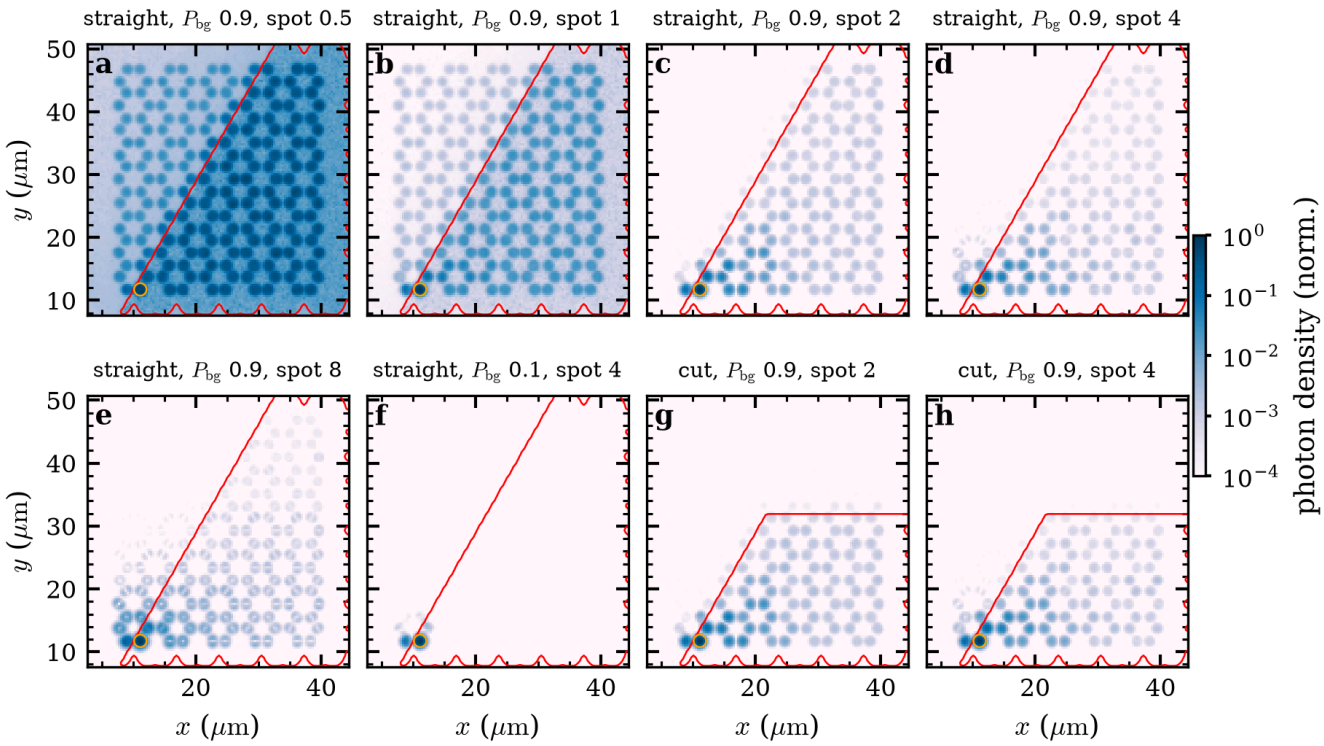


Figure S1. A small non-resonant spot lases locally but does not feed the interface. Time-averaged photon density (last 80 ps of 150 ps, log scale, each map normalised to its maximum) with an additional incoherent spot of 2 µm FWHM (orange circle, half maximum) on the injection pillar, for the background pumps and spot strengths given in the titles (units of P_{th}^{est}); red: pump edge. **a, b**, The weak spot is hidden in the emission of the large-area pump. **c–e, g, h**, The spot lases and its emission spreads into the pumped bulk around it. **f**, Without the large-area pump the emission stays on the spot. One seed, 350 nm grid.

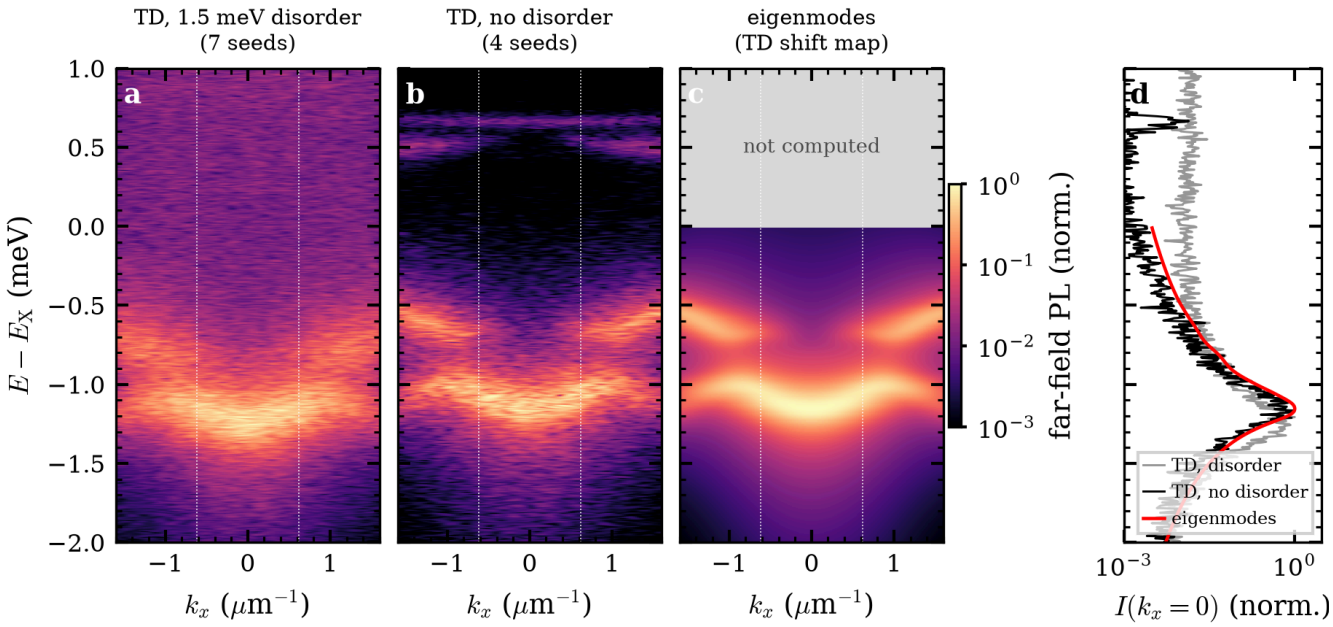


Figure S2. The time-dependent model reproduces the eigenmode band structure. Uniform incoherent pump at 0.3 $P_{\mathrm{th}}^{\mathrm{est}}$ on the whole lattice (no pump edge), 350 nm grid. **a, b**, Photoluminescence far field of the simulation with noise in the exciton and photon channels, slit k_y = 0, averaged over seven realisations with 1.5 meV exciton disorder (a) and four without (b); 819 ps record per realisation sampled every 0.02 ps, Hann window. **c**, Far field of the eigenmodes of the same lattice with the time-averaged exciton shift of the simulation, each mode weighted with its noise source ($0.1|X_m|^2\langle s\rangle_X + |C_m|^2\langle s\rangle_{ph}$, s: pump profile) divided by $\Gamma_{\mathrm{net},m}$ and broadened by $\Gamma_{\mathrm{net},m}$; no eigenmodes exist below −1.19 meV, modes above −0.01 meV were not computed (grey). **d**, Spectra at k_x = 0. Dotted: $k_x = \pm 4\pi/(3a_{\mathrm{latt}})$. Log scale, each panel normalised.

Reproduction (folder td_routing): model_td.py (model and parameters), 01_power_point.py and 02_analyze_power_sweep.py (Fig. 1), 03_routing_run.py (Figs. 2, 3, 6, round 5), 06_nonres_spot.py (Figs. 3d, S1), 10_uniform_farfield.py (pump v5) and 11_uniform_compare.py (Fig. 4a–d), 07_linear_response.py (Fig. 4e, f), 13_supercell_bands.py (Fig. 8), 08_edge_chain_ssh.py (Figs. 5, 6), 10_uniform_farfield.py and 11_uniform_compare.py (Fig. S2), 12_disorder_linewidth_test.py (disorder test, Sec. 4), report/plot_td_report.py (all figures and the animation). Data in td_routing/data.